IIT BOMBAY TECHFEST 2026–27 | ROUND 1 DELIVERABLE

NammaSpace

Interactive 3D Digital Twin & Real-Time Spatial Navigation Engine

Challenge: Namma Space
Round: 1 (3D Reconstruction & Roaming)
Status: Live Production Deployed

This official submission document provides direct access to the live, zero-install WebGL viewer and open-source code repository for the **NammaSpace** spatial digital twin challenge. All assets have been compiled and hosted for instant browser testing by the IIT Bombay Techfest evaluation panel.

🔗 OFFICIAL PROJECT ACCESS LINKS

🌐 1. Live Web Application (Zero-Install WebGL Viewer)

https://nammaspace-digital-twin.vercel.app/

Runs locked at **60+ FPS** in any modern desktop or mobile browser (Chrome, Edge, Safari, Brave). Features first-person exploration, dual-model switching (Mobile Scan vs. Ultra-HD Studio), interactive 3D waypoints, and real-time obstacle-avoiding floor pathfinding with a 2D synchronized radar.

💻 2. Open-Source Source Code & Pipeline Repository

https://github.com/Kaish-sheikh/nammaspace-digital-twin

Complete codebase containing the Three.js WebGL engine, 3D Gaussian Splatting scene loaders, Catmull-Rom pathfinding algorithms, 4-Pass smartphone capture Standard Operating Procedure (SOP), and sample trained splat binaries.

🎮 QUICK EVALUATION GUIDE FOR JUDGES

🕹️ Movement Controls
Use `W` / `A` / `S` / `D` to walk forward, left, backward, and right. Move your mouse to look around in first-person view.

🔄 Dual-Model Environment Switcher
Toggle between **Ultra-HD Studio (1.13M Splats, Primary Default)** and the **High-Density Scan (8.09 MB, 252K Splats)** via the top command header.

📍 Interactive 3D Waypoint Dock
Click any card in the left Waypoints dock to illuminate an obstacle-aware green laser walking path across the floor directly to that station.

📡 Real-Time 2D Floor Radar
Top-right radar minimap tracks your real-time heading with a rotating delta fighter chevron, FOV vision cone, and metric range rings.

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